DISCRETE STRUCTURES

CMSC250

Chapter 26: Equivalence Relations and Orders

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Equivalence classes, partitions, partial orders, and Hasse diagrams



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1 Equivalence relations

An *equivalence relation* on a set A is a relation that is reflexive, symmetric, and transitive. It formalizes when objects should be considered "the same" under a specified criterion, even if they are not literally equal.

2 A complete equivalence-relation proof

Question. On $\mathbb{Z}$, define $a \sim b$ if and only if $5 \mid (a - b)$. Prove that $\sim$ is an equivalence relation.

Proof. We verify the three properties.

1. **Reflexive.** Let $a \in \mathbb{Z}$ be arbitrary. Since $a - a = 0 = 5 \cdot 0$ and $0 \in \mathbb{Z}$, we have $5 \mid (a - a)$. Thus $a \sim a$.
2. **Symmetric.** Let $a, b \in \mathbb{Z}$ and assume $a \sim b$. Then $a - b = 5k$ for some $k \in \mathbb{Z}$. Hence $b - a = 5(-k)$; since $-k \in \mathbb{Z}$, $b \sim a$.
3. **Transitive.** Let $a, b, c \in \mathbb{Z}$ and assume $a \sim b$ and $b \sim c$. Then $a - b = 5k$ and $b - c = 5\ell$ for integers k, ℓ. Adding gives $a - c = 5(k + \ell)$; since $k + \ell \in \mathbb{Z}$, $a \sim c$.

Therefore $\sim$ is an equivalence relation. ■

Why the proof works. Each property is proved separately. Notice that the proof explicitly names arbitrary integers and invokes integer closure when producing a new witness.

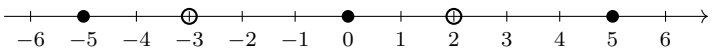
3 Equivalence classes

For an equivalence relation $\sim$ on A, the *equivalence class* of $a \in A$ is

$$[a] = \{x \in A : x \sim a\}.$$

For congruence modulo 5, $[0]$ contains $\ldots, -10, -5, 0, 5, 10, \ldots$, while $[2]$ contains $\ldots, -8, -3, 2, 7, 12, \ldots$.

Filled: [0]; circled: [2] (mod 5)



4 Classes are equal or disjoint

Question. Prove that if $[a] \cap [b] \neq \varnothing$, then $[a] = [b]$.

Proof. Suppose $[a] \cap [b] \neq \varnothing$. Choose $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$; by transitivity, $a \sim b$. Now let $x \in [a]$ be arbitrary. Then $x \sim a$, and since $a \sim b$, transitivity gives $x \sim b$. Thus $x \in [b]$, so $[a] \subseteq [b]$. Reversing the roles of a and b gives $[b] \subseteq [a]$. Hence $[a] = [b]$. ■

5 Partitions and equivalence relations

A *partition* of A is a family of nonempty subsets that are pairwise disjoint and whose union is A. Every equivalence relation partitions A into its distinct equivalence classes. Conversely, any partition defines an equivalence relation: two elements are related exactly when they lie in the same block.

Worked example: a partition defines equivalence

Question. Let $A = \{1, 2, 3, 4, 5\}$ be partitioned into $\{1, 3\}$, $\{2, 4\}$, and $\{5\}$. Describe the induced relation.

Solution. Two numbers are related if they occur in the same block. The relation consists of $(1, 1), (1, 3), (3, 1), (3, 3)$, the four ordered pairs among $\{2, 4\}$, and $(5, 5)$. It is reflexive, symmetric, and transitive because "in the same block" has all three properties.

6 Partial orders

A relation $\preceq$ on A is a *partial order* if it is reflexive, antisymmetric, and transitive. A set equipped with a partial order is a *partially ordered set* (poset). Two elements may be *incomparable*: neither $a \preceq b$ nor $b \preceq a$ holds.

Worked example: subset inclusion is a partial order

Question. Prove $\subseteq$ on $\mathcal{P}(A)$ is a partial order.

Proof. For any $S \subseteq A$, we have $S \subseteq S$, so reflexivity holds. If $S \subseteq T$ and $T \subseteq S$, then $S = T$, so antisymmetry holds. If $S \subseteq T$ and $T \subseteq U$, then every element of S lies in T and therefore in U, so $S \subseteq U$. Hence transitivity holds. Therefore $\subseteq$ is a partial order. ■

7 Total orders and comparability

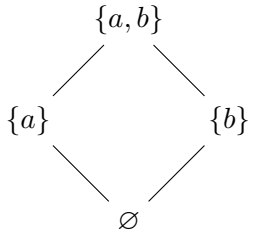
A partial order is a *total order* if every pair a, b is comparable: $a \preceq b$ or $b \preceq a$. The usual $\leq$ on $\mathbb{Z}$ is a total order. Subset inclusion on $\mathcal{P}(\{1, 2\})$ is not total: $\{1\}$ and $\{2\}$ are incomparable.

8 Hasse diagrams

A Hasse diagram represents a finite poset by drawing an upward edge for a *cover relation*; loops and edges implied by transitivity are omitted. Higher nodes represent larger elements in the order.

Worked example: power-set order

Question. Draw the Hasse diagram of $(\mathcal{P}(\{a, b\}), \subseteq)$.



Solution. The empty set is below both singletons; each singleton is below $\{a, b\}$. The singletons are incomparable, so no edge joins them.

9 Minimal, maximal, least, and greatest elements

An element m is *minimal* if no distinct element lies strictly below it. It is *least* if it lies below *every* element. Dually, maximal means no distinct element lies above it, while greatest means it lies above every element. A least or greatest element, if it exists, is unique; there may be several minimal or maximal elements.

Worked example: minimal is not necessarily least

Question. Consider $A = \{\{1\}, \{2\}, \{1, 2\}\}$ ordered by $\subseteq$. Find minimal, maximal, least, and greatest elements.

Solution. The minimal elements are $\{1\}$ and $\{2\}$ because neither has a smaller member of A. The unique maximal and greatest element is $\{1, 2\}$. There is no least element: neither singleton is contained in the other.

10 Bounds in posets

For a subset S of a poset, an *upper bound* u satisfies $s \preceq u$ for every $s \in S$. A *lower bound* satisfies $\ell \preceq s$ for every $s \in S$. A *least upper bound* is an upper bound below every other upper bound; a *greatest lower bound* is defined dually.

Worked example: bounds under divisibility

Question. On the positive integers ordered by divisibility, identify a greatest lower bound and least upper bound of $\{6, 10\}$.

Solution. Lower bounds are common divisors, so the greatest lower bound is $\gcd(6, 10) = 2$. Upper bounds are common multiples, so the least upper bound is $\operatorname{lcm}(6, 10) = 30$. Here "greatest" and "least" refer to divisibility order, not merely numerical size.

11 Practice questions and complete solutions

Question. 1. Is the relation $a \sim b$ iff a and b have the same parity an equivalence relation on $\mathbb{Z}$?

Proof. Reflexivity: $a - a = 0$ is even. Symmetry: if $a - b$ is even, then $b - a = -(a - b)$ is even. Transitivity: if $a - b$ and $b - c$ are even, their sum $a - c$ is even. Therefore it is an equivalence relation. Its classes are the even and odd integers. ■**Question.** 2. Is strict subset $\subsetneq$ a partial order on $\mathcal{P}(A)$?

Solution. No. A partial order must be reflexive, but $S \subsetneq S$ is false for every S. Hence strict subset is not a partial order. **Question.** 3. In the poset $\{2, 3, 6, 12\}$ under divisibility, find minimal and maximal elements.

Solution. Both 2 and 3 are minimal because neither is divisible by another distinct member of the set. The sole maximal (and greatest) element is 12. There is no least element because neither $2 \mid 3$ nor $3 \mid 2$. **Question.** 4. Prove that two distinct equivalence classes cannot share an element.

Proof. If they shared an element, the equal-or-disjoint theorem proved earlier would force them to be equal. Thus distinct classes have empty intersection. ■

12 Common mistakes

Equivalence relations require symmetry; partial orders require antisymmetry. Do not confuse minimal with least or maximal with greatest. A Hasse diagram omits self-loops and arrows implied by transitivity. When proving equivalence classes equal, show both inclusions or use the equal-or-disjoint theorem with its hypotheses.